

A Divide and Conquer Algorithm for Electron Microscopy Segmentation

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ABSTRACT

Cryo-Electron Microscopy is a biophysical technique able to visualize macromolecular complexes by producing 3-dimensional images. Currently, it has been advanced to be the second popular technique to construct protein molecules in terms of the number of structures released annually. The main advantages of cryo-electron microscopy are its ability to visualize large molecules, molecules that are hard to crystalize in their native environment. One critical step to construct the structure of a molecule from cryo-electron microscopy is to divide the image into regions for the chains/subunits that make up the molecule/complex. If the image is accurately segmented into the correct regions, the process of modelling using existing tools become easier and faster.

In this paper, we developed a divide-and-conquer algorithm to segment a given cryo-electron microscopy image efficiently. Our approach is based on the popular watershed algorithm. We tested our method on 10 authentic images and compared it with Segger. Although, it is difficult to conduct an accurate comparison, the results show that the performance of our algorithm is competitive when compared to Segger.

CCS CONCEPTS

• Applied computing→Life and medical sciences→Computational biology→Molecular structural biology • Applied computing→Life and medical sciences→Computational biology→Imaging • Theory of computation→Design and analysis of algorithms→Algorithm design techniques

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KEYWORDS

Cryo-Electron Microscopy, Density Map, Protein Modeling, Segmentation

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1 Introduction

Proteins are macromolecules composed of amino acids that are connected by peptide bonds. The interactions between amino acids groups and hydrophobic interactions in the polypeptide make up the tertiary structure of the protein. The biological function of a protein is directly relied on its structure. The biological functions proteins may perform are extensive and vary from structural, immunological, signal transporting, cell adhesion, cell cycle, and many more. Similar protein structure can perform similar functions [1].

Conventionally, protein structure is determined by experimental biophysical techniques. The main three biophysics techniques used to determine the structure of proteins are X-ray crystallography, nuclear magnetic resonance (NMR), and Cryogenic-Electron Microscopy (cryo-EM). To date, X-ray crystallography and NMR are the leading techniques for protein structure determination. However, they undergo numerous limitations [2, 3]. Further, they are not suitable for all types of proteins [4]. For instance, membrane proteins are difficult to determine by either X-ray or NMR. In general, the pace of determining protein structures using experimental techniques is relatively slow. Therefore, the gap of the number of proteins have associated structures and the number of proteins have associated sequences is increasing.

Cryogenic-Electron Microscopy (cryo-EM) is relatively newer than X-ray and NMR. It has been given large attention due to its potential. It is able to visualize and interpreting unstained nanostructures of biological complexes in their native state [4]. In contrast to the other two techniques, cryo-EM can visualize large molecules such as viruses or ribosomes and molecules that are hard to crystallize such as membrane proteins. Due to the recent advancement of technology and equipment, it has been ranked as the second popular technique to determine molecular structures after X-ray crystallography [5, 6].

Cryo-EM technique can produce 3-dimensional (3D) volumetric images at different resolutions. The resolution of these images ranges from high ($<5\text{\AA}$), medium ($5\text{\AA}$ to $10\text{\AA}$) to low ($>10\text{\AA}$). Recently, the number of cryo-EM images released manually increases due to improved detectors that are now available. At high resolution, most or entire part of the image is well visualized, and the subunits that make up the macromolecule are clearly visible and the separation of these units is clear. Consequently, the end-user can work on each individual segmentation/subunit to model the molecule. When the resolution degrades, the separation between the subunits is difficult to identify and the segments of the image belong to individual subunits are difficult to distinguish. Moreover, the manual segmentation and separation of the subunits for high resolution images is a time-consuming process.

In this paper, we propose a divide-and-conquer algorithm based on popular watershed algorithm to find the segmentation of cryo-EM images efficiently. The rest of the paper is organized as follows: Section 2 provides a summary of literature review and state-of-the-art techniques. Section 3 presents and discusses our proposed algorithm. Section 4, provides and discusses the experimental results. Finally, section 5 concludes the paper.

2 Related Works

The problem of image segmentation can be defined as the problem to divide a given 3D image into groups of voxels which are homogeneous based on a predetermined criterion. It can be defined as a labeling problem where voxels belong to the same homogeneous group are given the same label. Image segmentation is an essential practice to analyze the image structure. Numerous segmentation techniques have been developed in the last few decades. Some of its applications relate to many fields, such as medical imaging [7], document analysis [8], object recognition [9], and CCTV segmentation and quality inspection of materials [10]. Segmentation is important for cryo-EM images. Once the image is produced, it is used to model the molecule. Most of modeling techniques construct a single chain at a time. Therefore, a critical step is to segment the image into single chains or molecular subunits [11]. For instance, Figure 1 shows an example of a cryo-EM image constructed for Dengue virus [12]. In order to model the proteins on the outer shell, the image needs to be segmented first or the boundaries of each single protein molecule to be identified. Traditionally, segmentation of cryo-EM 3D image has been achieved by endless and biased manual process. Regardless of the level of resolution, the manual process to segment the image is

time-consuming. In general, it requires user professional knowledge in the domain such as knowledge in both protein structure and cryo-EM image analysis and processing. In addition, due to the presence of noise in cryo-EM images, the accuracy of the segmentation is not satisfied. There are several automatic and semi-automatic methods have been developed to segment the images. Some are based on watershed algorithm or elastic network. Watershed is a fast tool while sometime resulting in over segmentation. Elastic method is an automated segmentation method without interaction with user.

Various methods have been proposed to solve the problem of segmentation of cryo-EM images. Volkmann et al. [13] developed watershed transform algorithm. The algorithm starts at the voxel with a global density minimum and increases the intensity step with preset values. At each level of intensity, it checks the intensity of 26 neighborhood voxels. If the test voxel is neither in the cluster region nor the boundary region, it is assigned as a new region. After all voxels in the same intensity level are processed, it increases intensity with preset value as a new level then repeats the previous process until none voxel left. The results show that the algorithm is fast to process the segmentation. Further, it limits the user input parameters which saves users from tedious procedure. However, the parameter selection is still subjective and vague. There is still over segmentation in the results compared to experimental data.

Burger et al. [14] proposed a hierarchy of increasingly coarser map segmentation method based on Markov diffusion framework. The algorithm considers cryo-EM image as a weighted undirected graph. Each voxel with intensity in the image is represented as vertex in a graph, and its 26 neighborhood voxels are considered as edges. The weight of edges is a function of difference of intensity. Based on weights, a Markov transition matrix which contains probability of signal-travel between voxels is generated. The method outputs a hierarchy of segmentation of increasingly coarser map at different levels. The result shows meaningful sub-regions and gives the draft of segmentation expression to users for in-depth study with other methods.

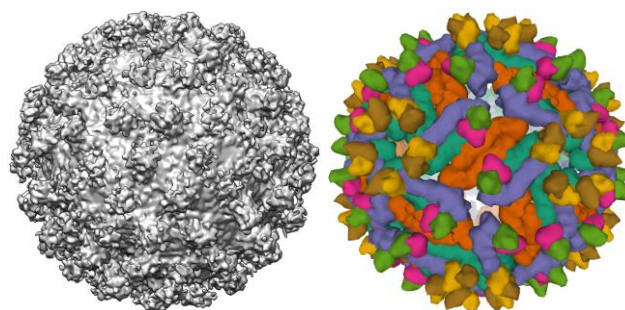


Figure 1: (left) Cryo-EM 3D image of “Dengue virus serotype 2 strain new Guinea-C” was produced at resolution 20 and shown at contour level 1.5 (EMDB ID: 2968). (right) coloring of the individual subunits/chains of the molecule.

Baker et al. [15] employed a modified fast-marching approach, called volRover, which includes three major steps: detection of critical voxels, detection of existence of global or local symmetry

axes and detection of boundaries of subunits. The approach suffers from high computational cost. MAINMASTseg is one of the latest tools that uses a graph structure to segment the image by considering symmetry corresponding intensity points in the graph [11]. One of the most popular tool is Segger [16]. Segger is based on watershed method. It iteratively smooths cryo-EM image by gaussian filter and then divides the image based on local maximum regions. It applies the process of iteratively grouping of the regions to reach the desired final number of segments. Segger is available as a plugin in the popular visualization tool and framework, UCSF Chimera [17].

3 Methodology

In this paper, we propose a highly efficient segmentation algorithm. We show the methodology in three steps. First, we present a preliminary algorithm. It uses watershed method. Then, we propose a new segmentation algorithm using divide and conquer technique. Instead of using conventional binary divide and conquer, we use square root divide and conquer to maximize the efficiency. We show that the proposed algorithm is extremely fast. Finally, we improve our new algorithm to prevent the result falling to a local optimal solution. In the improved algorithm, we identify a group of special voxels evenly distributed in the cryo-EM image and add them in the merge steps of the divide and conquer algorithm.

3.1 Preliminary Algorithm

The preliminary segmentation algorithm consists of two phases. The first phase of the algorithm finds a set of preliminary regions/segments by watershed method. It preprocesses the volume 3D image to remove noise, i.e., remove the voxels if their density/intensity values are less than the threshold. The threshold is the value based on visual inspection, which required user to input at this stage. Then it finds the preliminary regions by assigning each voxel p to a region r if r has the largest local maximum (the voxel with the largest density in the region) among the regions that p 's neighbors belong. The output of the first part of the algorithm includes a set of preliminary regions and a set of preliminary local maxima, where local maxima are the voxels with the largest density in each region. Notice that the size of the preliminary regions achieved in the first phase is small and the number of regions is large. The second phase of the algorithm repeatedly merges the regions until the whole segmentation process completes. A region A is merged to a region B if the steepest ascent of local maximum in region A is the local maximum in B . The detailed algorithm is given as follows:

Algorithm Preliminary Segmentation – Phase I: Initialization of the segments using Watershed

Input: D – density map

Output: R_0 – a set of preliminary regions, M_0 – a set of corresponding preliminary local maxima

Step 1 (Smooth & filter)

Sort D in decreasing order of density value, then remove the values lower than threshold.

Step 2 (Initialization of the segments)

for each density value d in D in decreasing order do

if voxel v is not adjacent to any voxel in the existing regions, v is assigned to a new region r , r is added to R_0 , v is added to M_0 ;

else v is adjacent to the voxel/voxels in the existing region/regions

if v is adjacent to the voxel/voxels in an existing r' in R_0 , v is assigned to the region r' ;

if v is adjacent to the voxels in multiple regions, v is assigned to the region r' with the largest number of adjacent voxels

3.1.1 Analysis of Preliminary Segmentation – Phase I. Let n be the number of voxels in D . Step 1 is dominated by sorting in $O(n \log n)$ time. In Step 2, each voxel v in D is processed once. For assigning v to a region, v has to find which region that most v 's neighbors belong to. Therefore, Step 2 can be completed in cn' time, where $c = 28$ (number of v 's neighbors), and n' is the number of voxels after filtering which is smaller than n .

Algorithm Preliminary Segmentation – Phase II: Merge of regions

Input: D, R_0, M_0, K (K is the number of final regions)

Output: Segmented regions

$M = |M_0|$; $R = R_0$;

Sort M in increasing order of density;

repeat the following steps **until** the number of regions $\leq K$

for each element x in the first half of M

find x 's steepest ascent y in M (x 's steepest

ascent is located in the elements after x in M), merge the

region of y , say r_y , to the region of x , say r_x , where r_y

and r_x belong to R .

The steepest ascent of v is defined as

$C \times \max_{u \in M} \frac{\text{density}(u) - \text{density}(v)}{\text{distance}(u, v)}$, where C is the parameter in the normalization for balancing the density and distance of pixels. In the implementation, we set $C = \left(\frac{\max_{u, v \in M} (\text{distance}(u, v))^2}{\max_{u, v \in M} (\text{density}(u, v))} \right)^{\frac{2}{3}}$

3.1.2 Analysis of Preliminary Segmentation – Phase II. In the algorithm, the size of M is the number of regions at each repeat step. At the beginning it is $|M_0|$, the size of the set of preliminary local maxima. The size of M reduces a half in each repeated merge step. It is easily to understand that if x is the i th element in M , x 's steepest ascent is located in the element after $(i+1)$ th element. Let $m = |M_0|$. The time complexity of Phase II is calculated as follows:

Time complexity of Phase II =

$((m-1) + (m-2) + \dots + (m/2)) + ((m/2-1) + (m/2-2) + \dots + (m/4)) + ((m/4-1) + (m/4-2) + \dots + (m/8)) + \dots + K = O(m^2 \log m)$. Usually, m is much smaller than n .

Theorem 1. The time complexity of Preliminary Segmentation algorithm is $O(n \log n + (m^2 \log m))$, where m is the number of preliminary regions obtained in Phase I of the algorithm.

3.2 Square-Root Divide and Conquer Segmentation

In Preliminary Segmentation algorithm, Phase II requires $O(m^2 \log m)$ time. Though m is much smaller than n , say m is 10% of n , it is still very large. The segmentation is not efficient enough for large proteins. In this section, we propose a new segmentation algorithm using square root divide and conquer.

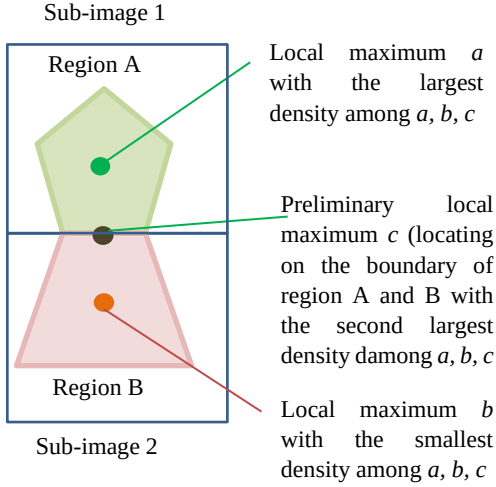


Figure 2 Region B doesn't merge to region A if voxel c belongs to region A only: a is too far away from b to be the steepest ascent of b .

Algorithm Square-Root Divide-and-Conquer Segmentation (D)

Input: D – density map, K – number of final regions

Output: Segmented regions

if $|D| (= n)$ is small enough, segment D using Phase I of the Preliminary Segmentation algorithm which returns a set of preliminary regions and a set of corresponding preliminary local maxima;

else

Step 1: Evenly divide D to $\sqrt{n}$ sub-images $D_1, D_2, \dots, D_{\sqrt{n}}$, such that each of them has $\sqrt{n}$ voxels.

Step 2: Call Square-Root Divide-and-Conquer Segmentation (D_i) recursively for each i from 1 to $\sqrt{n}$, where each will output K segmented regions.

Step 3: Use Phase II of the Preliminary Segmentation algorithm to merge the regions of $\sqrt{n}$ sub regions obtained in Step 2 to K regions.

3.2.1 Analysis of Square-Root Divide-and-Conquer Segmentation. In the algorithm, each of $\sqrt{n}$ sub-images produces K regions; therefore, the total number of regions for the merge in Step 3 is $K\sqrt{n}$. The time complexity of Step 3 is the same as that of Phase II in Preliminary Segmentation algorithm, where $m = K\sqrt{n}$. The recurrence of the time complexity for the algorithm is given as follows:

$$T(n) = \sqrt{n}T(\sqrt{n}) + c(K\sqrt{n})^2 \log(Kn), \text{ where } c \text{ is a constant}$$

$$= n^{\sum_{i=1}^h \frac{1}{2^i}} T\left(n^{\frac{1}{2^h}}\right) + cK^2n (\log(Kn^{1/2^{h-1}}) + \log(Kn^{1/2^{h-2}}) + \dots + \log(Kn))$$

Let $h = \log \log n$, $T(n) = c'n + cK^2n \log \log K + cK^2n \log n$, where c' and c are constants.

Theorem 2. The time complexity of Square-Root Divide and Conquer algorithm is $O(K^2n \log n) = O(n \log n)$.

Theorem 2 shows that the proposed square root divide and conquer algorithm largely improved the time complexity of the preliminary one.

3.3 Improved Square-Root DC Algorithm

In the proposed divide and conquer algorithm, sometimes the segments are not merged as they should. It happens when the voxels on the boundary of sub-images are not used properly in the merge of regions. In the divide and conquer algorithm, when a voxel on a boundary is assigned to one sub-image, it is not for other sub-images which share the same boundary. It causes the problem sometimes. To show the problem, let us see Figure 2. In the figure, voxel a is the local maximum in region A , voxel b is the local maximum in region B , voxel c is located on the boundary of A and B but is assigned to A , where $\text{density}(b) < \text{density}(c) < \text{density}(a)$. Recall that in the merge process a region is merged to its steepest ascent. In Figure 2, Region B may not be merged to region A because a is too far away from b to be the steepest ascent of b . However, if we assign voxel c to both regions of A and B , then c will be the local maximum of region B ; furthermore, region B will merge to region A because a is close enough to be c 's steepest ascent (see Figure 3).

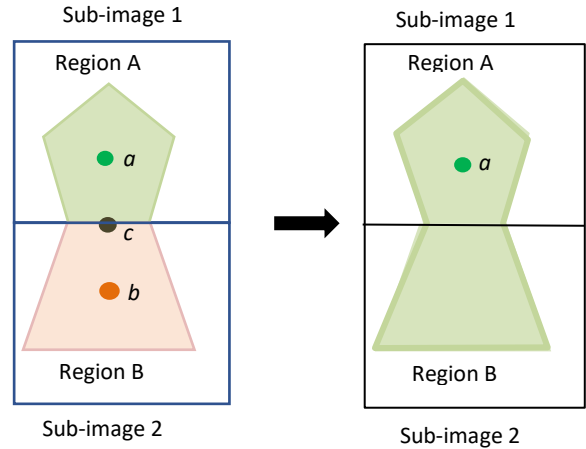


Figure 3: Region B merges to region A if voxel c belongs to both regions A and B : c becomes the local maxima of B and a becomes the steepest ascent of c

Based on the above observation, we know that the voxels on the boundary of sub-images, especially, those with relatively large density, should join the region merge process. In the improved the algorithm, we add the preliminary local maximum into the local maxima if they are on the boundary of the sub-images in the region merge process.

4 Experimental Results

We used a group of 10 authentic cryo-EM images to compare between our method and one of the most popular existing tools, Segger. The cryo-EM maps are randomly chosen from EMDataBank (EMDB) from various resolution levels (between 2.2 Å -17 Å) [18]. In order to use the image in the testing, it should have a corresponding protein model. The structure models are downloaded from Protein Data Bank (PDB) online [19]. In order to find the accuracy of the tools, we used the chains of the protein as the reference cuts to compare with. A masked region is created that corresponds to protein's chain. To create the masked region (i.e., reference cryo-EM segmentation), the authentic image is superimposed with each chain of protein molecule. The density within a radius of 2 Å around every atom of the protein is saved into the reference cut. These cuts are then used as reference cuts to compare between Segger and our own method.

Segmentation accuracy is quantitatively calculated using the method proposed in [16]. The method is called shape-match score (SM) as follows:

$$SM = \text{volume}(R \cap T) / \text{volume}(R \cup T)$$

Where R is the reference cut and T is the segment produced by the tool, either by Segger or our method. The score is calculated at voxel level. For each segmented region using the tool, the score is calculated with the reference cut and the accuracy is reported. A score is a single numerical value in the range [0,1]. A high score means an accurate segmentation. The score is 0 when two regions do not match at all and it is 1 when the two regions are identical.

Segger is implemented as an interactive plug-in that comes with UCSF Chimera. It uses an iterative watershed method to segment a given cryo-EM image. Segger requires some parameter input to operate. To compare against our method, we ran Segger using some of the recommended parameters by its authors and the tutorial they have posted. One of the parameters Segger requires is an initial contour threshold to start the segmentation. We used the same level of contour for Segger and our algorithm (see Table 1). We used a threshold that is good in such a way that the components are visually clear to the user. The other parameters we used for Segger are as follows: smoothing steps: 4, step size: 1, Keep only regions having at least 10 voxels. We use the default value for the rest of parameters.

EMDB ID	PDB ID	resolution	Contour level	# chains	Segger			Proposed Method		
					# segments	average accuracy %	best chain accuracy	# segments	average accuracy %	best chain accuracy %
3444	5M5C	4.8	0.02	5	4	56.3	61	5	41.2	47
7962	6O9G	4.6	0.03	4	4	22.5	29	3	31.0	39
8438	5TQY	5.2	0.04	2	2	80.0	82	2	80.0	80
9950	6K9L	4.27	0.01	2	2	41.5	54	2	84.0	84
10574	6TTE	2.2	0.06	4	4	94.0	94	3	83.0	83
21121	6V9I	5.2	0.4	2	2	44.5	52	2	42.0	70
21533	6W2X	3.6	0.01	2	2	18.5	25	2	50.0	50
2557	4D2U	17	0.25	6	6	26.2	28	6	33.5	35
3435	5G4G	7.8	0.06	6	6	49.3	60	5	57.0	66
0505	6NT8	6.5	0.013	4	4	33.8	47	4	31.0	54
Average				3.7	3.6	46.7	53.2	3.4	53.3	60.8

Table 1: The performance of Segger and our proposed method using 10 authentic cryo-EM images

In Segger, the final number of segments for a given cryo-EM image is not directly provided by the user. The user controls the number of segmented regions by pressing the “Group” button. In general, Segger first segments an input image into many small pieces of regions. The regions are iteratively merged into larger and fewer regions when users press the “Group” button. To optimize a segmentation result for an image, we grouped segmented regions manually by pressing the Group button until the number of regions became equal to or smaller than the number of chains in the image. For our method, the user provides the initial contour threshold and the number of final regions to generate. The

method will generate several regions that are equal or smaller than the provided number.

For a segmented region, the SM score is calculated against all reference cuts. The segmented region will be assigned to the reference cut if it has the largest SM with. Table 1 shows the results of the two methods. In this study, we are not comparing between the two methods to rank them based on the performance. The goal is to characterize the performance of our algorithm when compared to Segger. It is difficult to conduct an accurate comparison because each method has a set of input information and have different development strategies or requires user interaction which is

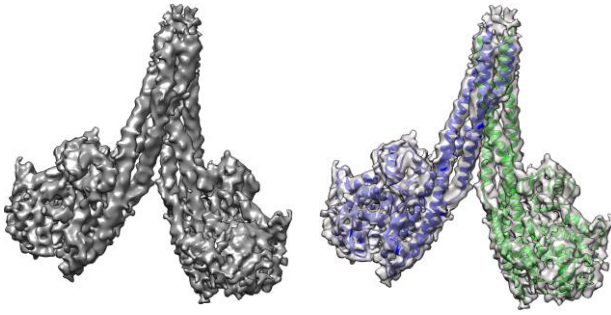


Figure 4: The authentic cryo-EM reconstruction of human IKK1 (EMDB ID: 8438). Left: the original cryo-EM image. Right: the chains of protein molecule (PDB ID: 5TQY) are superimposed.

relative to user level of experience. The evaluation was performed in terms of SM and number of regions detected by each method.

Table 1 shows that our method has performed comparatively to Segger. As we see from table 1, the performance of our method is close to the performance of Segger. When Segger performs well in segmenting the image, our method also performs well. For example, in both images, 8438 and 10574 (EMDB ID), both methods could find most of the regions with accuracy, on average, more than 80%. Figures 4-6 show the image of 8438 (EMDB ID) and the regions segmented by the two methods. On the other hand,

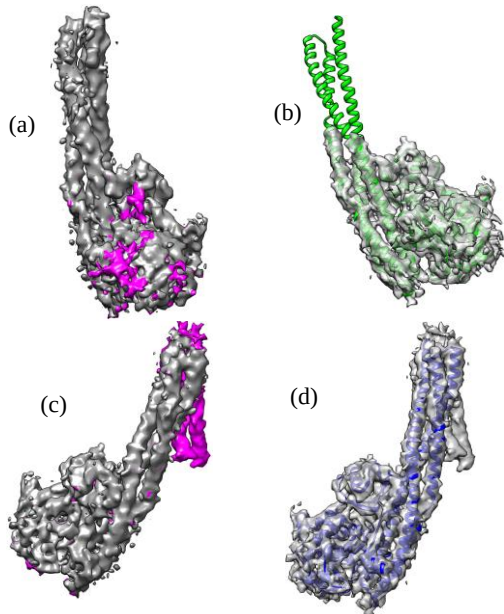


Figure 5: The segmentation results for cryo-EM reconstruction of human IKK1 (EMDB ID: 8438) using Segger. (a) The segmented region is superimposed with the reference cut (gray) for chain A. (b) The segmented region produced by Segger is superimposed with chain A structure. (c) The segmented region is superimposed with reference cut (gray) for chain B. (d) The segmented region is superimposed with chain B structure.

Segger performs slightly better than our method in finding most of reference regions. When both methods compared, we find that

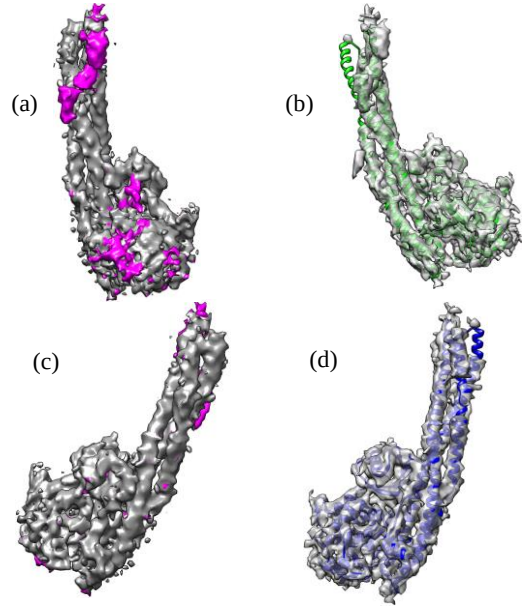


Figure 6: The segmentation results for cryo-EM reconstruction of human IKK1 (EMDB ID: 8438) using our method. (a) The segmented region is superimposed with the reference cut (gray) for chain A. (b) The segmented region produced by Segger is superimposed with chain A structure. (c) The segmented region is superimposed with reference cut (gray) for chain B. (d) The segmented region is superimposed with chain B structure.

Segger was able to find the correct number of regions in nine (9) out of the 10 images. In contrast, our method was able to find the correct number of regions in seven (7) out of the 10 images. In terms of the accuracy, the average of SM scores for the regions segmented by our algorithm is slightly better than Segger. Note that the SM score for 80% of the regions is less than 60% for both methods. This is due to the nature of images picked for the test and it is expected for authentic images.

5 Conclusion

The biological function of a protein is tightly connected to its 3D structure. Undoubtedly, knowing the structure of any protein is indispensable to understand its function. However, the pace of determining protein structures using conventional biophysics techniques is slow. Due to advanced technologies nowadays, many cryo-EM images have been resolved to different resolution levels. Computational tools that are used to model protein molecules using cryo-EM construct single chain/subunit at a time. Therefore, the segmentation of a given cryo-EM image into regions that are corresponding to the molecules making up the target macromolecule is crucial.

In this paper, we proposed an efficient divide and conquer algorithm based on watershed to segment a given cryo-EM image. We presented a preliminary algorithm that is improved by a square root divide and conquer technique. Although the comparison between two not fully automatic tools are inaccurate, we compared the performance of our method with Segger. The test was performed on 10 authentic images. The results show that our method was competitively equivalent to Segger. However, more analysis will be conducted to improve the results so the number of regions to segment is more accurate.

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